IE 692: Optimization Methods for Reinforcement Learning

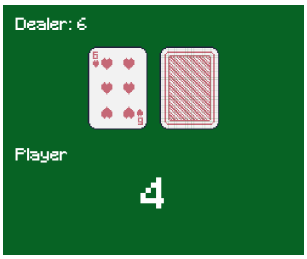
Course Project

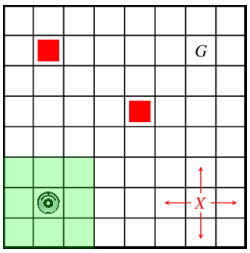
Presentation+code: Nov 24th (8am EST) Report: Nov 24th (11:59pm EST)

v1 is the original file. Red text are changes for v2, and blue text are changes for v3.

Problem Description

You and your team are tasked with implementing a complete and basic reinforcement learning (RL) library from scratch. The team members are expected to apply their knowledge from the first three units – formulations and models, policy optimization, and policy evaluation – to solve practical control and decision-making problems.





All projects must consist of the following three components:

1. Model and solve for two finite state and finite action Markov decision processes in the *online model*: a simplified blackjack game and GridWorld (more details given in the *Details* section);
2. Implement a Truncated On-Policy Monte Carlo (TOMC) with dynamic exploration time for Policy Evaluation. More details about tuning given in the *Details* section;
3. Implement a tuned Stochastic Policy Mirror Descent (SPMD) for Policy Optimization;

In addition to these standard implementations, each team must investigate three enhancements to either policy evaluation or policy optimization. These enhancements are to be compared with the standard implementations above to uncover more numerically promising methods.

The list below is a non-exhaustive list of possible enhancements (for material not covered in class, a reference is given):

- Policy evaluation: importance sampling OMC, TD learning (with a tuned step size/learning rate, mixing length, and function approximation), OMC with a dynamic exploration time[1], ...
- Policy optimization: SPMD with three different Bregman divergences (Euclidean distance squared, KL-divergence, and Tsallis divergence), Stochastic Policy Dual Averaging with three different Bregman divergences[2], Block SPMD[3], compare against PPO (we recommend the CleanRL implementation[4]) ...

[1] Section 3.2 in https://arxiv.org/pdf/2512.06244.
[2] Algorithm 3 (Section 4) in https://arxiv.org/pdf/2211.16715
[3] Algorithm 2 (Section 4) in https://arxiv.org/pdf/2201.05756.
[4] https://github.com/vwxyzjn/cleanrl/blob/master/cleanrl/ppo.py

Students are expected to use their own experience and judgment on how to best implement these methods. Team members are encouraged to be creative and suggest their own enhancements or test on different environments. While feedback will be provided after the project proposal, students are encouraged to discuss with the instructor about their own enhancements to gauge its difficulty and scope.

The project outcomes include:

1. Formulate and model a decision-making problem as a Markov decision process;
2. Gain familiarity with basic implementations of policy evaluation and policy optimization methods and empirical enhancements;
3. Exposure to tuning to RL methods to attain better empirical results;
4. Apply the scientific method to design, test, and analyze numerical experiments.

Submission Instructions

- For the proposal, please upload (only one team member):
    1. A single **typed** PDF with the team members and outlining the three proposed enhancements.
    2. Please confirm that all team members are available to present on Tuesday, November 24th from 12:55pm to 2:10pm EST. Distance learning students can present remotely while non-distance learning students are expected to be in-person. If there are potential conflicts, please contact the instructor ASAP.
- For the final project, please upload (only one team member):
    1. A single **typed** PDF report (must include title, authors, a short executive summary/abstract, and main sections as described in *Details* below) and a PDF copy of the group presentation.
    2. Python code. Please provide clear instructions on how to run your code (and installation steps for external libraries). And have your code be readable (e.g., avoid large functions, good variable naming, write succinct comments). Hard-to-read code may lose points.

Details

- Given the current class size, each group have 3 students, with exceptions allowed for 2 with instructor approval. If there are any group conflicts that cannot be solved, please privately email the instructor as soon as possible.
- Out of 100pts: 10pts (proposal), 50pts (project design and execution), 40pts (written and oral presentation). See *Rubric* below.
- Some enhancements will not be covered until later in the semester. Please look ahead in the text, use online resources, and discuss with the instructor to learn more for a head start.
- Your RL code must handle *online data*.

- Further details about MDP environments: For simplicity, choose a discount factor of $\gamma = 0.9$. The simplified BlackJack game is based on the Gymansium framework[5], where the deck only has cards with numerical value 2-5 (drawn from an infinite deck with replacement), the dealer will draw cards until their sum is 9 or greater, and a bust occurs when the sum of the cards exceeds 11. The GridWorld environment is based on Lecture 1, but it must be enlarged to be a 10x10 grid with 10 hazards, 3 goal states, and 1 origin/starting state. If you find your method cannot all visit all states in GridWorld within a reasonable time frame, feel free to adjust the environment and algorithm as you see fit. If you do so, please justify your modifications in your final presentation and report.

- Further details about project design and execution: your code must measure the performance of each implemented algorithm by 1) estimated state-value function $V^{\pi_t}(s)$ over each iteration t (use your best judgment for a reasonable state to evaluate at) and 2) number of observations made in each iteration t. For tuning algorithmic parameters, choose up to three important algorithmic parameters, and consider three reasonable values for each. Then apply a grid search, where each combination is evaluated by running the RL method with a limited observation budget (e.g., 5-10% of the total budget of a full run). Select the set of parameter combinations that yields the best performance for the full run.

- Further details on written and oral presentation: Your written report must include an introduction (describe the proposed enhancements and hypotheses for expected behavior), a methodology section (describe each enhancement at a high-level and provide 1-3 high-level details about implementation details and their rationale), a results and analysis section (a visually appealing plot for the value and observation progression, as well as explanations that support the observed results, compare against the hypothesis, and detail possible reasons for the (un)expected behavior), and a conclusion summarizing the key findings and future directions. Also include an addition section describing the contribution of all members.

- Correctness of results and code will be based on course lectures and assignment problems. Please contact the instructor for any questions about code design or results early on.

- In general and unless otherwise explicitly stated by the instructor, the submitted report and code represents the group's collective efforts, and therefore all group members will receive the same grade and be subject to AI usage violations as a group. Group members are encouraged to work together and check all submitted reports and code for potential issues. If individual members face any conflicts or have any concerns with other members that are beyond their control, please contact the instructor early for alternative grading criteria.

Rubric

In extreme cases, scores may be adjusted based on group member contribution (see the last bullet point above).

Course Project Grading Rubric

[5] https://gymnasium.farama.org/environments/toy_text/blackjack/.

Criterion	90-100%	80-90%	60-80%	0-60%
Proposal (10pts)	Complete proposal	-	-	Incomplete or missing
Design (16.6pts)	3 distinct and appropriate enhancements	Limited scope in 1 enhancement	Minor overlap or limited scope in 2-3 enhancements	Major overlap or inappropriate scope
Code (16.6pts)	Clear and succinct code	Clear high-level code, small number of major issues	Working code that is hard to read	Broken code or non-matching results.
Results (16.6pts)	Reasonable and/or explainable results with correct plots	Correct results with 1-2 unexplained phenomena and/or plot errors	Plausible results with 3+ unexplained phenomena and/or plot errors	Incorrect and/or unexplained results and plots
Written Report (20pts)	Correct format; clear and concise	Correct format; 1-2 major confusion points	Slightly incorrect format; 3+ major confusion points	Incorrect format and poorly written report
Oral Presentation (20pts)	Clear and concise presentation; thoughtful responses	Mostly clear and concise, with 1-2 major confusion points that were unanswered	Acceptable presentation, multiple confusion points that were unanswered	Unclear or disorganized presentation

Examples of good code: short-to-medium-length functions, good variable naming, succinct and clear comments, good layout. **Avoid unnecessary functions, complicated folder structures, and code designs.**

Technical requirements:

- Code: See under "Details." Correctness of code is based on model design, reproducibility, and SPMD (TOMC with dynamic exploration time. KL-divergence is OK; see Assignment #2).
- Written Report: See under "Details". See also Argonne's Research Report (link here). Plots must be included, as well as any instructions (as needed) to reproduce the results.
- Oral Report: Maximum 10min with additional 1-2min of questions. Must address the project's methods, results, discussions, and team members' contributions.

Bonus: Up to 10pts additional can be earned for suggesting a practical MDP that is validated with real-world data.

Course policy and submission details

Your submission must be of original work and effort. AI technology and online references are allowed and must be properly cited. Submissions that are very similar to a reference, include an AI link/screenshot without showing the prompts and generated output, or do not appear to be of original work and effort (at the discretion of the instructor) will be considered for plagiarism. Collaborations are allowed and must include the name of other student(s) and high-level description of shared discussions that impacted the submission. Submission can directly use material from office hours and lectures without needing to cite. The final report has a one hour grace period submission extension, but the presentation and code will not.

Presentation Schedule

	Members	Time
Group 1	Ray, Vetter, Wu	12:55-1:05
Group 2	Basher, Mahmud, Mashtura	1:07-1:17
Group 3	Ali, Banerjee, Chowdhury	1:20-1:30
Group 4	Zhang, Zhao	1:32-1:42
Group 5	Budathoki, Dhakah, Morris	1:45-1:55
Group 6	Clark, Crews, Mishra	1:58-2:08